Problem 1: TPD Analysis

Thermal desorption of CO from Rh(111) ; 10 K/s ; Coverage dependent desorption

Redhead equation: $E_{des} = RT_{max}\left[\ln\left(\frac{\nu T_{max}}{\beta}\right) - 3.46\right]$; $\beta = 10\ K/s$; $R = 0.0082\ kJ/(mol\ K)$

(a)

T_{max} / K	$10^{13}\ s^{-1}$	$10^{14}\ s^{-1}$	$10^{15}\ s^{-1}$
510	127.16	136.78	146.41
400	98.93	106.49	114.04
300	73.49	79.16	84.82

↳ Activation energies in kJ/mol

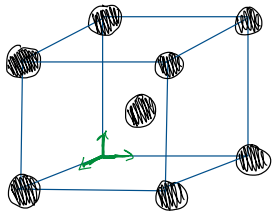
(b) They would be equal if we assume no interactions b/w adsorbate species ↗ reversibility b/w ads and des.

(c) The preexponential factor is chosen such that it is independent of coverage.
A choice of higher values indicate that there is a greater change in entropy b/w the molecule in ground state & transition state.

Problem 2: Crystal Structures

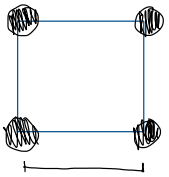
BCC ; Rhombic dodecahedral crystal shape

(i)



Body centered Cubic

→ 100 plane



$a = \frac{4R}{\sqrt{3}}$; R = radius of Fe atoms

↑ unit cell length

0.126 nm

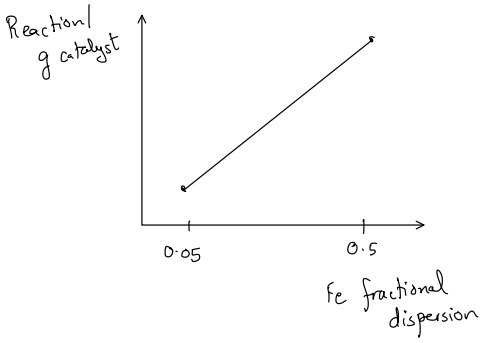
$a = \frac{4}{\sqrt{3}}(0.126 \times 10^{-7})\ cm =$

Exposed atoms per cm^2 : $\frac{1}{a^2} =$

For (110) Fe : $\frac{1}{\left(\frac{4R}{\sqrt{3}}\right)^2\ cm^2} = \frac{1}{\left(\frac{4 \cdot (0.126 \times 10^{-7})}{\sqrt{3}}\right)^2}\ \frac{atoms}{cm^2} =$ $\boxed{1.18 \times 10^{15}\ \frac{atoms}{cm^2}}$

(ii) CO hydrogenation ≠ f(Fe crystal size)

Fe dispersion varied from 0.05 → 0.50 } This results in more (exposed) surface Fe atoms to which CO can react w/



$$D_m = \frac{N_s}{N_T} = \frac{[\text{surface area}]\,(cm^2)\,(\rho_s)\,\frac{\text{atoms}}{cm^2}}{[\text{volume}]\,(cm^3)\,(\rho_v)\,\frac{\text{atoms}}{cm^3}} = \frac{8\sqrt{2}\,r^2}{\frac{16\sqrt{3}}{9}\,r^3}\cdot\frac{\frac{1}{r^2}}{\frac{1}{a^2}}$$

$$D_m = \frac{8\sqrt{2}}{2}\cdot\frac{9}{16\sqrt{3}\,r}\cdot\frac{a}{a}$$

edge length of dodecahedral

$$r = \frac{9\sqrt{2}}{4\sqrt{3}}\,\frac{a}{D_m}$$

$r = 1.07 \times 10^{-6}$ cm

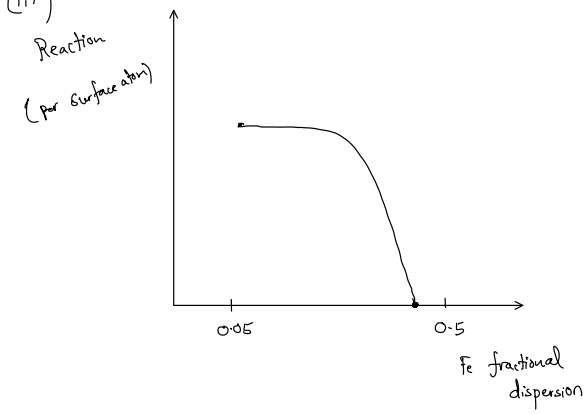
$= 1.07 \times 10^{-6} \times 10^{-2}$ m

$= 1.07 \times 10^{-8}$ m

$1.07\times10^{-7}\times10^{-2}$

	r = 10.7 nm	@	D_m = 0.05	
&	r = 1.07 nm	@	D_m = 0.5	→ diameter range will be twice of this assuming $D = 2r$

(iii)

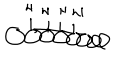


NH_3 reaction on surface atoms

↓

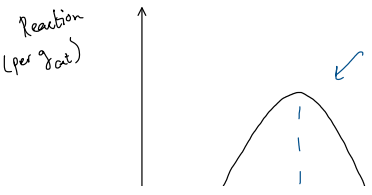
As dispersion ↑, surface atoms ↑

reaction rate accounts for increase in surface atoms

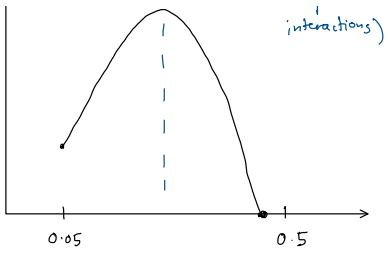


However, the rate decreases over time as the 4N sites are saturated → so not all dispersed sites will be used.

(especially as SA↓ so many sites are just blocked)



← Max rate where 4N adsorbate clusters are spread out (minimizing interactions)



(iv) H_2 chemisorption is not applicable on Fe since H_2 exhibits activated adsorption on Fe and thus will not give accurate results [Vanice]

Dispersion on Fe catalyst can be measured via CO chemisorption. CO adsorbs non-dissociatively but can adsorb linearly on single atom site or at a bridge where it binds to 2 sites

↳ Requires 2 isotherms for uptake for reversible uptake

CO chemisorption:

1) CO introduced and CO uptake ($\mu mol/g_{cat.}$) is measured as function of pressure
2) If support is present then blank runs w/ support are also measured
3) Reversible & irreversible uptake are deconvoluted
   - ↳ weak adsorption from support (Henry's law regime @ high pressures) ↳ Low coverage proportional to adsorbate pressure
   - strong chemisorption on reduced metal surface (Langmuirian isotherm → high pressure regime)
4) dispersion is calculated using total no. of atoms (metal loading) & the surface sites information from CO uptake
5) Average crystallite size is then calculated as $D = \left(\frac{0.6}{d}\right)\left(\frac{V_m}{A_m}\right)$ where for spherical particle assumption [Vanice]

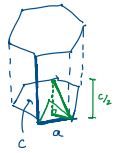
   D = dispersion, d = crystallite size, V_m = bulk volume density, A_m = bulk area density

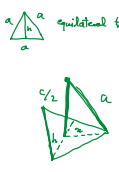
For TEM:

Given a histogram of # of particles w/ given size → the average can be calculated via a binomial distribution fit on the dataset.

Surface averaged & number-averaged crystallite size would be similar for materials where surface exposure is not reflective of its activity for chemisorption i.e. non-metallic sites or single atom sites

(v) Ideal HCP axis size ratio





equilateral triangle: $h = \sqrt{a^2 - \left(\frac{a}{2}\right)^2} = a\sqrt{\frac{3}{4}}$

$x = \frac{2}{3}h = \frac{1}{\sqrt{3}}a$

$\therefore \frac{c}{2} = \sqrt{a^2 - x^2}$

$= \sqrt{a^2 - \frac{1}{3}a^2}$

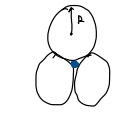
$\frac{c}{2} = a\sqrt{\frac{2}{3}}$

$$\frac{c}{a} = \sqrt{\frac{8}{3}}$$

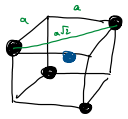
Interstitial sites exist b/w the spaces w/n a unit cell.

Closed Cubic Cell (CC) has { 4 octahedral sites (w/ coordinats to 6 atoms) & 8 tetrahedral sites (w/ " to 4 atoms)

HCP has { 6 octahedral sites 1/
12 tetrahedral sites

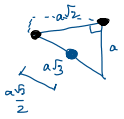


● → Tetrahedral site w/ radius r



$2R = a\sqrt{2}$

$R = \frac{a\sqrt{2}}{2}$



$(R+r) = \frac{a\sqrt{3}}{2}$

$\therefore \frac{a\sqrt{2}}{2} + r = \frac{a\sqrt{3}}{2}$

$r = \frac{a(\sqrt{3}-\sqrt{2})}{2}$

$r = 0.159\,a$

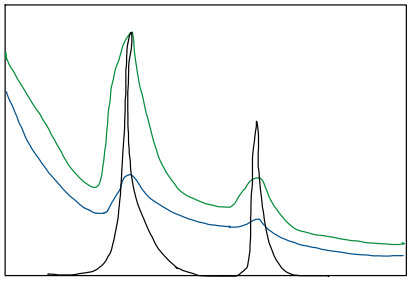
or $\boxed{r = 0.225\,R}$

For an atom to occupy a tetrahedral void, its radius must be 0.225 times the radius of Fe

HCP and FCC crystal structure are quite similar [(where the FCC(111) plane corresponds to the closed packed plane of HCP (001))] } Therefore the voids are similar

Problem 3: XRD Analysis

(a)



— Pd reference
— 2.4 wt%. Pd loading on silica
— 1.1 wt%. Pd loading on silica

+ Reference signals are higher intensity & lower FWHM since Pd is a crystalline sample & will diffract x-rays effectively

+ SiO_2 is amorphous and therefore lower wt loadings will give less crystalline (more amorphous signals)

(b) (111) plane:

$n\lambda = 2d\sin\theta$; $2\theta = 40.2°$
$\lambda = 0.154$ nm for Cu Kα
n = order of reflection ($n=1$)

$$d = \frac{(0.154\text{ nm})}{2\cdot\sin(40.2°/2)}$$

$d_{111} = 0.224\ nm$

(c) Pd lattice constant => 0.389 nm ; Pd is simple cubic

$$\left(\frac{1}{d_{hkl}}\right)^2 = \frac{h^2+k^2+l^2}{a^2}$$

$$\left(\frac{1}{d_{111}}\right)^2 = \frac{1^2+1^2+1^2}{a^2}$$

Solving for a^2 → $a^2 = 3\cdot(d_{111})^2$

$a = \sqrt{3}\cdot d_{111}$

$a = 0.388\ nm$ ✓ which is almost equal to 0.389 nm

(d) If using $M_o\ K\alpha$ → $\lambda = 0.07\ nm$

For d_{111} : $n\lambda = 2d\sin\theta$

$$\frac{0.07}{2}\cdot\frac{1}{d_{111}} = \sin\theta$$

$\theta = 8.8°$

$2\theta = 17.98°$ for (111)

Similarly, d_{200} :

$$2\theta = \sin^{-1}\left(\frac{0.07}{(2.)(0.194)}\right)\cdot 2$$

$2\theta = 20.79°$ for (200)

(e) $\langle L\rangle = \frac{K\lambda}{\beta\cos\theta}$ → Shemer equation

According to the Shemer equation → measuring at smaller wavelengths gives sharper peaks since λ is smaller and diffraction lines shift to lower angles decreasing the $1/\cos\theta$ term (as shown in (d))

↳ Both effects reducing line broadening → and smaller particles can be probed } more resolved peaks

Question 4

H_2 chemisorption on 1.5% Rh/Al_2O_3 catalyst

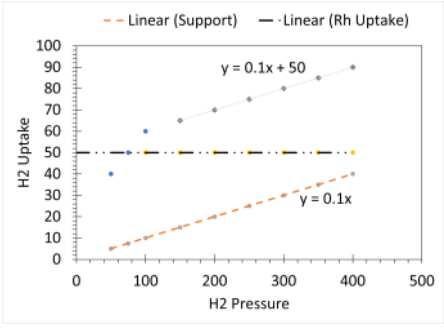
$$H_2 + 2Rh_s \rightarrow 2\ Rh_s - H$$

(Rh_s: Rhuthenium surface atoms)

Assuming Rh does not form bulk hydride phase: we can separate the graph into 2 parts

1) Henry's Law regime → weak reversible adsorption on support
2) Adsorption on only Rh crystallites

H2 Pressure (Torr)	H2 Uptake (umol/g_cat)	Support uptake	Difference ("a")
50	40	5	35
75	50	7.5	42.5
100	60	10	50
150	65	15	50
200	70	20	50
250	75	25	50
300	80	30	50
350	85	35	50
400	90	40	50



y-intercept → gives the saturation adsorption on Rh crystals

$50\ \mu mol / g\text{-catalyst}$

slope → henry law slope w/ intercept of 0 } uptake by support only

Difference is H_2 uptake by Rh → $50\ \mu mol / g_{cat}$

Coverage:

$1\ mol\ H_2 \rightarrow 6.022 \times 10^{23}$ atoms $\rightarrow 2$ H atoms

H_2 uptake for $1\ g_{cat} \rightarrow 50 \times 10^{-6}\ mol \cdot \left(2 \times 6.022 \times 10^{23}\right) \frac{atoms}{mol} = 6.022 \times 10^{19}$ atoms

$$Rh_s \Rightarrow 6.022 \times 10^{19}\ atoms / g_{cat}$$

Dispersion = $\frac{N_s}{N}$ (N_s: surface atoms; N: total no. of atoms)

$= \frac{Rh_s}{Rh_T} = \frac{6.022 \times 10^{19}\ atoms}{(8.778 \times 10^{19}\ atoms)} =$

$D = 0.69$

$1\ g_{cat} \rightarrow$ has 1.5% Rh $\rightarrow$ 0.015 g of Rh

$\downarrow$

$MW(Rh) = 102.91\ g/mol$

$\downarrow$

1.458×10^{-4} mol of Rh

$\downarrow$

$Rh_T = 8.778 \times 10^{19}$ atoms

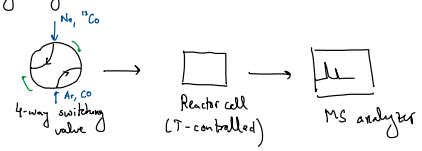
Question 5

Method development for measure SA of Pt catalyst during WGS reaction via steady state isotopic transient kinetic analysis (SSITKA)

WGS reaction: $CO + H_2O \rightleftharpoons CO_2 + H_2$

Theory: Detecting time scale of evolution of labelled products & correlating w/ evolution of labelled surface active species (via time-resolved FTIR)

Gas delivery system:



→ We know CO adsorbs on Pt (usually linearly) → therefore we can calculate total moles of CO adsorbed on Pt by $^{12}CO/^{13}CO$ isotope switching to perform CO oxidation to get total surface area of catalyst

→ However not all surface sites are "active" for WGS reaction and therefore we can get idea about active metal sites via istopic transient measurements as such:

(1) Isotope tracer (Ne) & ^{13}CO labelled gas is introduced into reactor cell via the 4-way switching valve

(2) The mean surface residence time of ^{13}CO is calculated by integrating the area b/w Ne trace (which elutes first) and $^{13}CO_2$ (the WGS product)

(3) The surface conc. of Carbon-containing active intermediates is then calculated as the product of the rate & the mean surface residence time (which can be corrected for any support influences)

(4) This value can then be compared w/ the # of total surface atoms (via H_2 chemisorption). Assuming a certain no. of surface sites required to bind w/ the intermediate (i.e. one for CO binding w/ Pt via top site) an estimate of "active" surface area for WGS on catalyst can be determined.

Question 6

Surface Area measurement via BET;

BET equation:

$$\theta = \frac{N}{N_m} = \frac{C\,(p/p_o)}{\left(1-\frac{p}{p_o}\right)\left(1-\frac{p}{p_o}+C\left(\frac{p}{p_o}\right)\right)}$$

; C = BET constant; p_o = vapor pressure of liq. N_2; p = equilibrium pressure; N = Adsorbed gas [mol/g]; N_m = Monolayer of adsorbed gas [mol/g] (theoretical)

↓ Linearized

$$\frac{(p/p_o)}{v\left[1-\frac{p}{p_o}\right]} = \frac{(C-1)}{v_m C}\left(\frac{p}{p_o}\right) + \frac{1}{v_m C}$$

; v = adsorbed gas quantity (mass)

BET plot: $\frac{1}{v}\left[\frac{p/p_o}{1-p/p_o}\right]$ vs. $\frac{p}{p_o}$ (for $0.05 < \frac{p}{p_o} < 0.35$)

Cell Volume:

$PV = nRT$; No. of moles in calibrated volume; $n = \frac{PV}{RT} = \frac{P_i \cdot (1801\ cm^3)}{RT}$

Assuming He does not adsorbed to sample → no. of mols of He should be equal to mol in calibrated volume + mol in cell

↳ Moles in cell build up since it is sequential pumping

↳ Assuming there is no heat transfer b/w cell reservoir & calibrate volume manifold we can get estimate of cell volume

↳ Assuming there is no heat transfer b/w cell reservoir & Calibrate volume manifold we can get estimate of cell volume

N_2 adsorbed :

Amount present @ initial + Amount introduced in cell volume = Amount adsorbed + Amount remaining (final)

Amount adsorbed = (Amount present @ initial + Amount introduced in cell volume) − Amount remaining

w/ sequential pumping: initial of N_2 are the no. of moles in manifold.

Amount adsorbed by sample = Total in Cell − remaining in cell

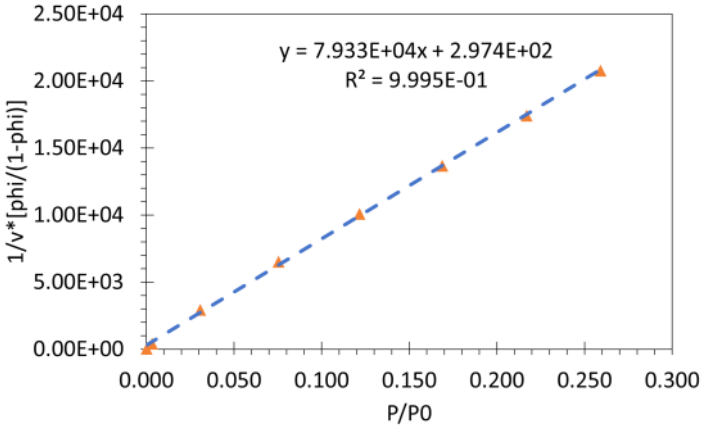
(N_2 introduced (by pressure increase) in Calibrated volume + N_2 remaining in the cell from previous run + N_2 remaining in the calibrated volume after previous run − N_2 remaining after end of current run in calib. volume)

Constants		
R	62.364	L Torr/(mol K)
	62364	cm3 Torr /(mol K)
T	21.2	C
	294.35	K
T_cell	77	K
Calib. Volume	18.01	cm3
P_0 (N2 @77K)	101.3	kPa
	759.8	Torr
Ax (ads. cross section)	0.162	nm2
	1.62E-19	m2
MW N2	28.0	g/mol
N_Av	6.02E+23	atoms/mol
Molar Volume (N2)	2.24E+01	dm3/mol
	2.24E-02	m3/mol
Mass of sample	2.60E-01	g

Raw Data		Calculated Values			
P_i (Torr)	**P_f (Torr)**	**n_He_i (mol)**	**n_He_f (mol)**	**n_cell (mol)**	**Cell volume (cm3)**
19.6	4.8	1.92E-05	4.71E-06	1.45E-05	14.53
51.8	16.5	5.08E-05	1.62E-05	4.92E-05	14.31
68.4	29.4	6.71E-05	2.88E-05	8.74E-05	14.28
79.4	41.9	7.79E-05	4.11E-05	1.24E-04	14.24
			Average	14.34	cm3
			Total Volume	**32.35**	**cm3**
			(Calib. Volume + Cell volume)		

Raw Data		Calculated Values											
P_i (Torr)	P_f (Torr)	N2 @initial (mol)	N2 @final_calib. vol (mol)	N2 introduced (mol)	N2 @final_cell volume (mol)	Initial (additive)	N2_total @cell (mol)	N2 adsorbed (cell) (mol)	N2 adsorbed cum.	Phi = P/P0	1/v (m3 @STP)	(1/v)*[(P/P0)/(1-(P/P0)]	*Check: (1/v)*[P/(P0-P)]*
162.3	0.1	0.0001592334	0.0000000981	0.0001592334	0.0000002985	0.0001592334	0.0001591353	0.0001588368	1.59E-04	0.000	2.81E+05	3.70E+01	*3.70E+01*
207.5	2.5	0.0002035794	0.0000024528	0.0002034813	0.0000074636	0.0002035794	0.0002014252	0.0001939615	3.53E-04	0.003	1.27E+05	4.18E+02	*4.18E+02*
220.2	23.4	0.0002160394	0.0000229579	0.0002135867	0.0000698596	0.0002160394	0.0002005452	0.0001306856	4.83E-04	0.031	9.23E+04	2.93E+03	*2.93E+03*
235.9	57.3	0.0002314428	0.0000562173	0.0002084849	0.0001710664	0.0002314428	0.0002450850	0.0000740186	5.58E-04	0.075	8.01E+04	6.53E+03	*6.53E+03*
254.9	92.4	0.0002500838	0.0000906541	0.0001938664	0.0002758558	0.0002500838	0.0003304961	0.0000546403	6.12E-04	0.122	7.29E+04	1.01E+04	*1.01E+04*
289.4	128.3	0.0002839319	0.0001258758	0.0001932778	0.0003830335	0.0002839319	0.0004339119	0.0000508784	6.63E-04	0.169	6.73E+04	1.37E+04	*1.37E+04*
324.3	164.9	0.0003181725	0.0001617843	0.0001922967	0.0004923011	0.0003181725	0.0005394217	0.0000471207	7.10E-04	0.217	6.29E+04	1.74E+04	*1.74E+04*
336.6	196.9	0.0003302401	0.0001931797	0.0001684558	0.0005878356	0.0003302401	0.0006293615	0.0000415260	7.52E-04	0.259	5.94E+04	2.08E+04	*2.08E+04*

BET Graph calcs		
slope	7.93E+04	
Intercept	2.97E+02	
v_m	1.26E-05	m3
Surface Area	5.47E+01	m2
Specific SA	**210.43**	**m2/g**



From the BET plot:

Slope = s ; intercept = I

$v_m = \frac{1}{s+I}$; $C = 1 + \frac{s}{I}$

S_{total} = Total surface area = $\frac{v_m N_A A_x}{V}$

A_x: adsorption cross section of adsorbate (common value of $0.162\ nm^2$ is used for N_2)

V: molar volume of adsorbate gas; for N_2: $22.393\ \frac{dm^3}{mol} = 22.393 \times 10^{-3}\ \frac{m^3}{mol}$

$m^3 \cdot \frac{atom}{mol} \cdot \frac{mol}{m^3} \cdot \frac{m^2}{atom} [=] m^2$

Specific SA = 210.43 m^2/g